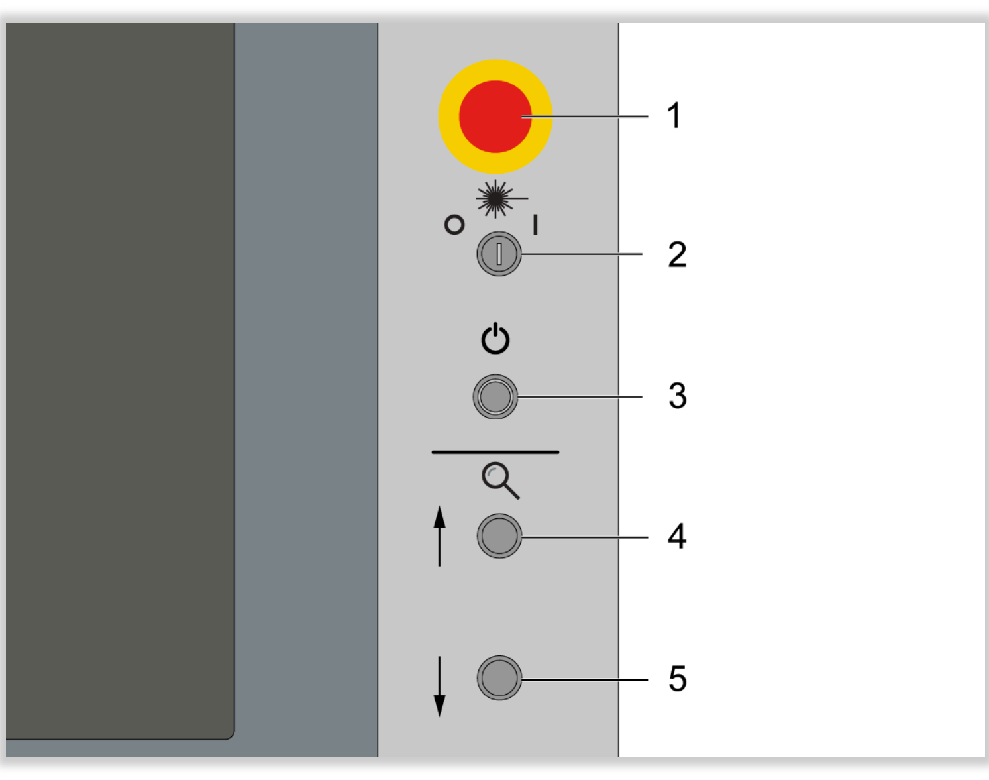
Introduction

This procedure will show you how to operate and maintain the Haas Laser - 50E/XT Series Machine.

Pre-Operation

Before operating the HL-50E/XT, it is important to become familiar with the machine's controls and complete the necessary pre-operation steps. This section explains the function of each control button and provides the information needed to prepare and safely operate the Haas Laser Marker.



[1] **EMERGENCY STOP**: The laser will be shut down immediately.

[2] **LASER ENABLED**: Gives power to the laser, light, and the galvonmeter.

[3] **LASER POWER**: Turns on the laser and galvonmeter.

[4] **LENS UP**: Motorized lift raise up.

[5] **LENS DOWN**: Motorized lift descend down.

Machine Power-On/Off

This section tells you how to power-on and power-off the laser marker.

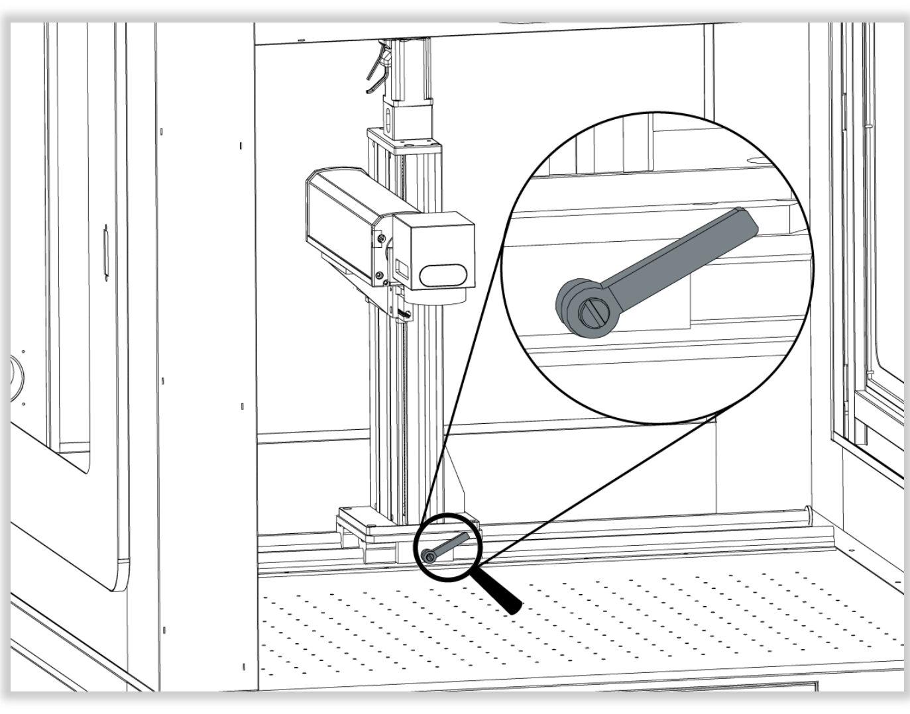
⚠ **IMPORTANT**: There is a specific order sequence to turn on the laser in order for the laser to turn on when "LASER POWER" is pressed.

1. Make sure the power cable connector is connected to the laser and to an outlet.
2. Breaker switch is on.
3. E-Stop is released
4. LASER ENABLED is keyed to "ON"
5. LASER POWER is pressed.

⚠ **IMPORTANT**: There is also a sequence to turn off the computer.

1. Laser must be connected to power
2. Breaker switch is ON
3. Computer switch is pressed to ON
4. Monitor cables are connected
5. Monitor is ON

X-Axis Adjustability



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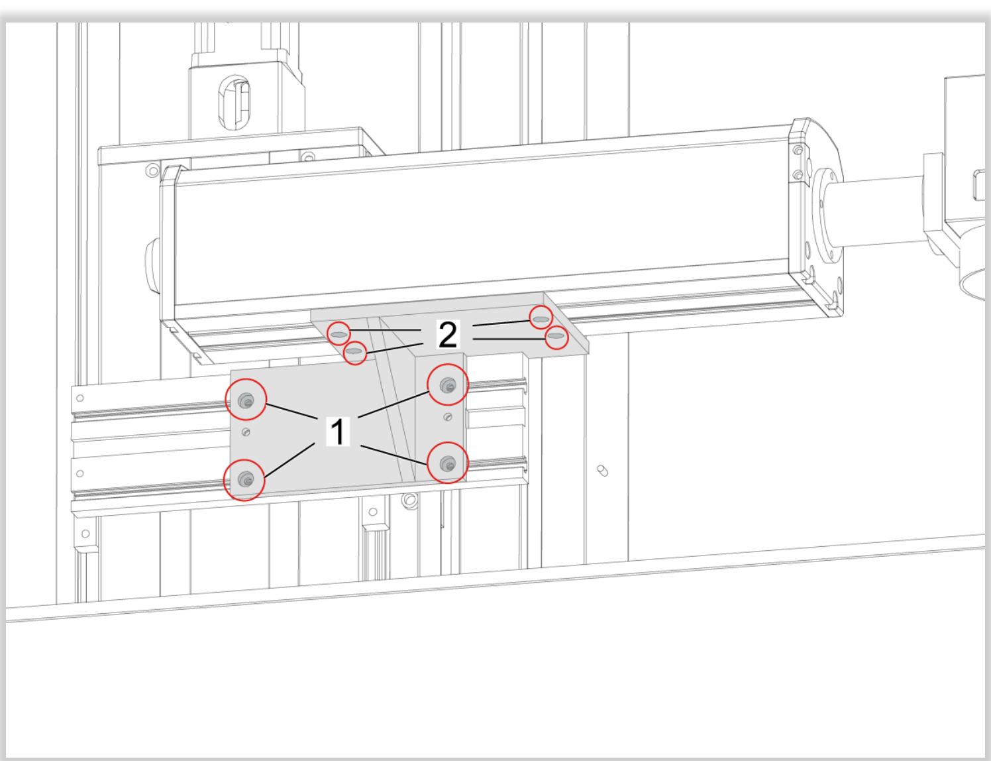
HL-50E-XT ONLY

Loosen the X-axis handle screw [1] to allow the laser head to move side to side along the X-axis. Before running a project, tighten the screw as this acts as a stop to keep the laser head from moving during operation.

⚠ **WARNING: Do not actuate or displace the laser head along the X-axis during active laser operation, as this poses significant safety risks and may result in equipment damage or personal injury. Ensure that all side access panels and the main enclosure door remain securely closed while the laser is in use to maintain proper containment and interlock functionality.**

Y-Axis Adjustability



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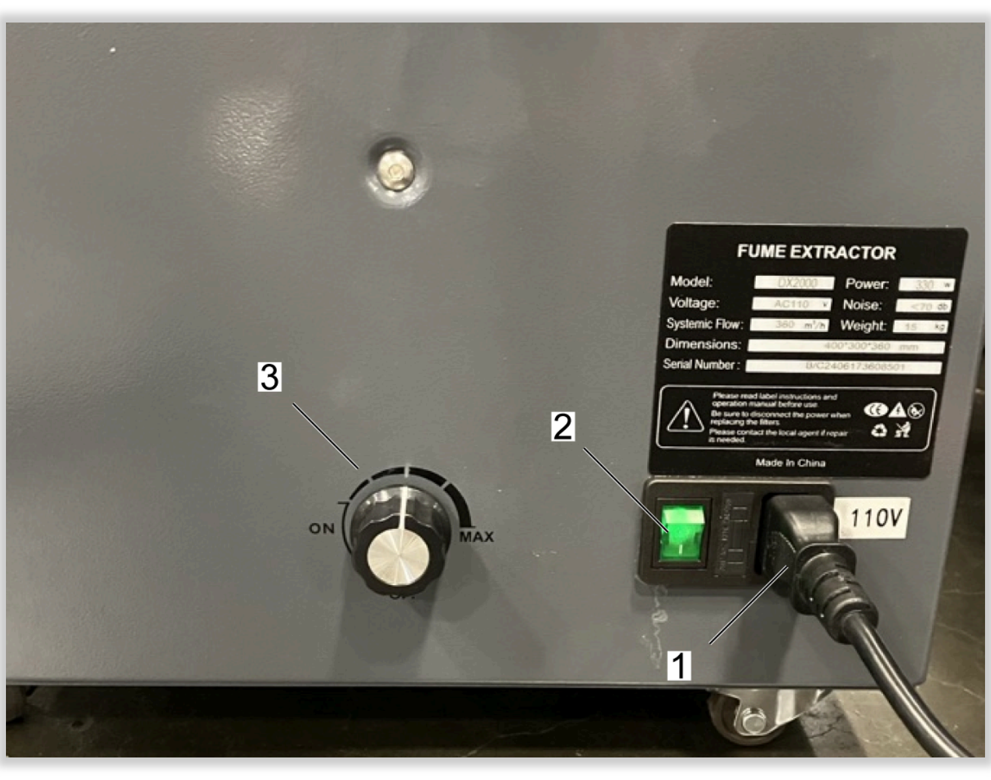
HL-50-XT ONLY

1. Loosen the set screws marked [1] to enable the laser head to move along the Y-axis.
2. If additional travel is required, proceed to loosen the set screws marked [2] for further adjustment.
3. After making the necessary adjustments, be sure to tighten all set screws.

Fume Extractor Power-On



To turn fume extractor ON, connect the female side of the power cable [1] to the fume extractor and connect the male side to an outlet.

Use switch [2] to turn the fume extractor on or off.

The fume extractor knob [3] can be used to adjust the intensity of the vacuum.

To completely turn off the fume extractor, move the knob counter-clockwise (CCW) past the **ON** label until you hear the click.

Getting Started with the HL-50E/XT

The HL-50E uses LightBurn as its user interface. Once LightBurn is activated, the laser is ready to operate. To learn more on how to set up your first project, refer to FIRST LIGHTBURN PROJECT DOCUMENTATION.

Swapping Lenses

⚠ **Important**: The HL-50E/XT comes equipped with two different F-Theta Laser Lenses:

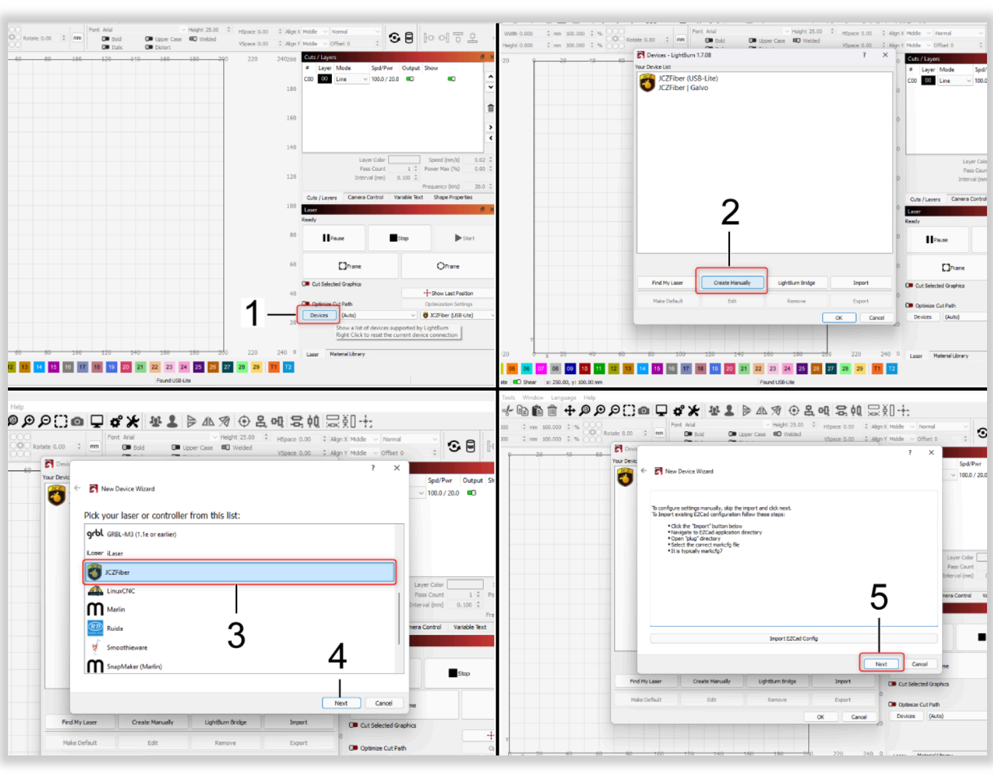
- F160 Lens: 110x110mm (~4x4") working area, 160mm focal length *(default)*
  - ~390mm (15.4in) maximum part height
- F290 Lens: 200x200mm (~8x8") working area, 290mm focal length
  - ~240mm (9.4in) maximum part height

Note: Laser comes with a F160 lens installed and already has a saved device profile for this lens in the LightBurn Software. Be sure to remove the lens protective caps from the lens before operating the laser.

The F290 lens does not have a device profile in LightBurn. You will need to create a device profile for the F290 lens in LightBurn. LightBurn will then save the configurations for the lens in that device profile and will allow you to switch lenses based on the requirements of your project.

Tools Required:

✓ Dark Colored Construction Paper (Black is recommended for best results)
✓ Caliper



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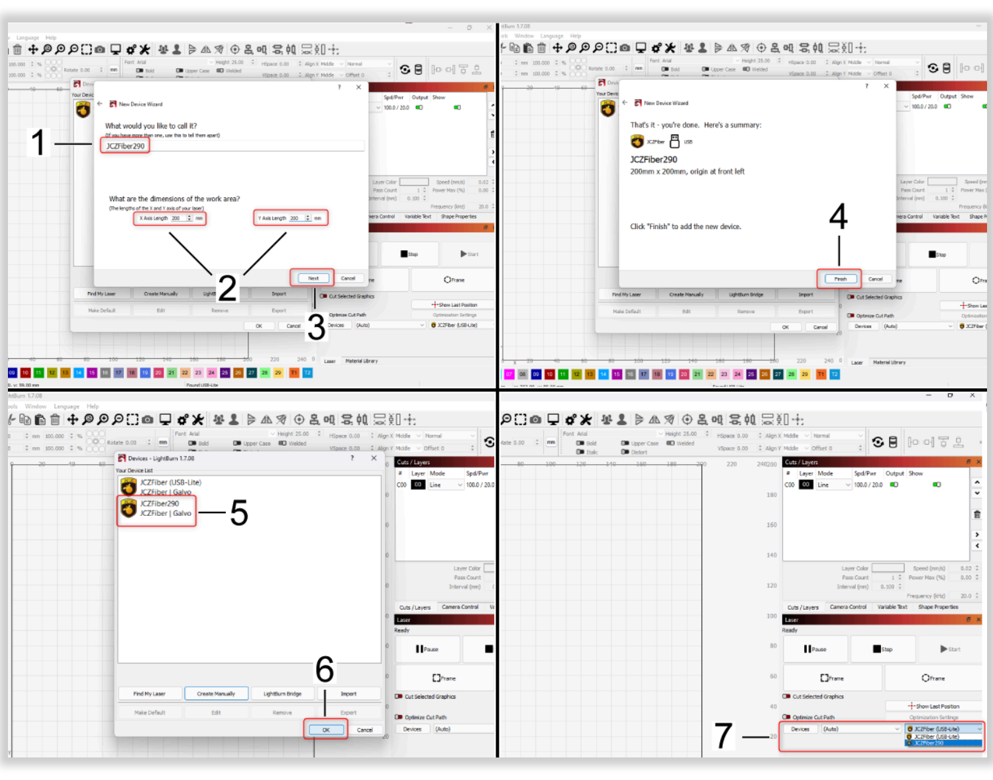
To swap the lens to the F290 F-THETA LENS, you will need to do the following:

1. Select Devices
2. Create Manually
3. Select JCZFiber
4. Click Next
5. Click Next



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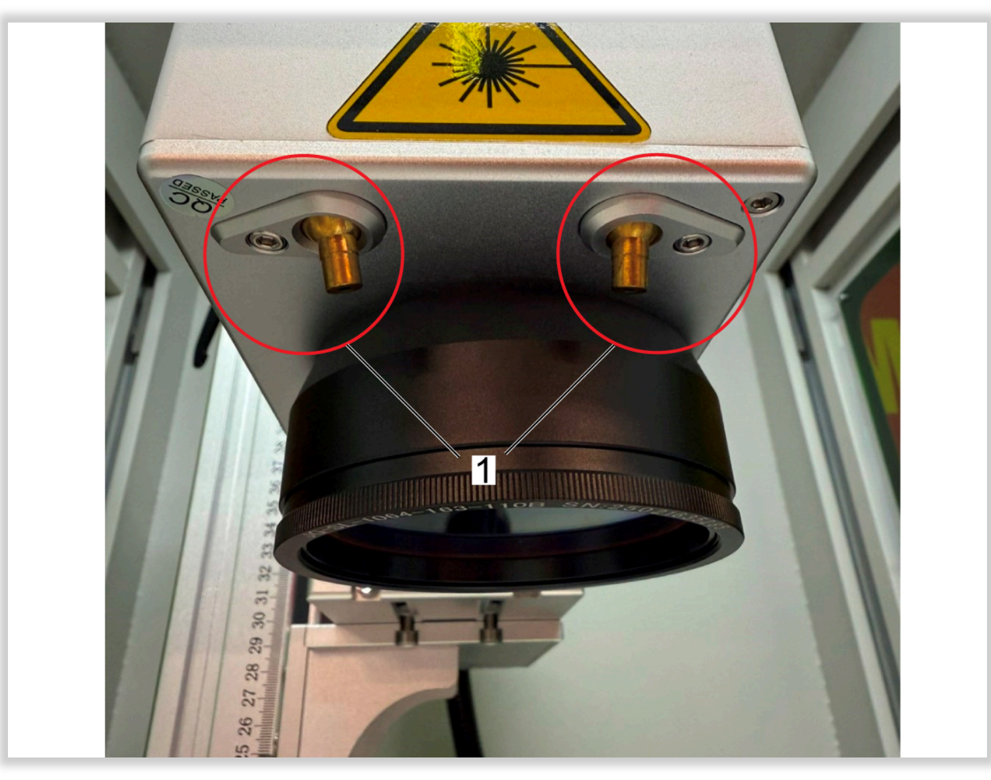
Swapping Lens instructions continued:

1. Name your laser. We recommend naming it based on the focal length of the lens. Such like JCZ Fiber290
2. X-axis Length and Y-Axis Length depends on the work area of the lens. For the F290 lens, input the following values:
   X-Axis: 200mm
   Y-Axis: 200mm
3. Click Next
4. Check that the Name and the Work Area of the lens is correct and click Finish if it looks ok.
5. Select the F290 in the device list
6. Click OK
7. In the Laser Window, you can see the lenses saved in your device folder. You can select the appropriate lens for your projects in the Laser Window.



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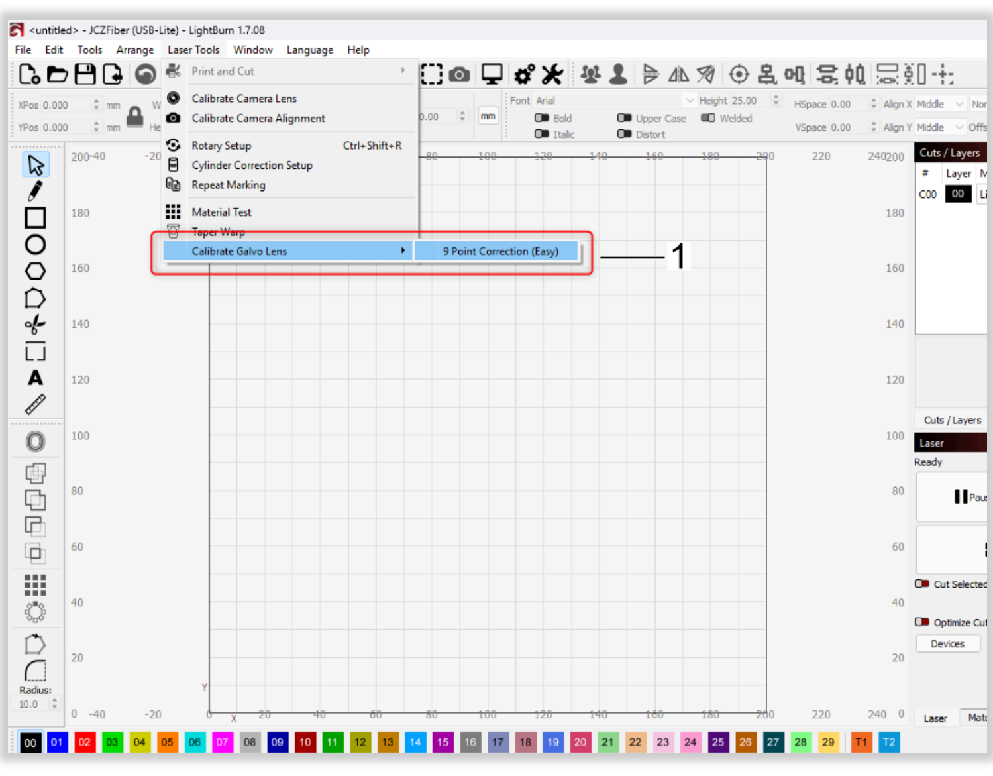
⚠ **Important**: Set your focal point before calibrating the lens. Refer to the GALVO LASER FOCUSING - LIGHTBURN DOCUMENTATION for instructions on how to set your focal point for the lens.

Once the focal point for the lens has been determined, manually adjust the set screws [1] until the two red diode dots converge. This sets the new focal point for the installed lens.



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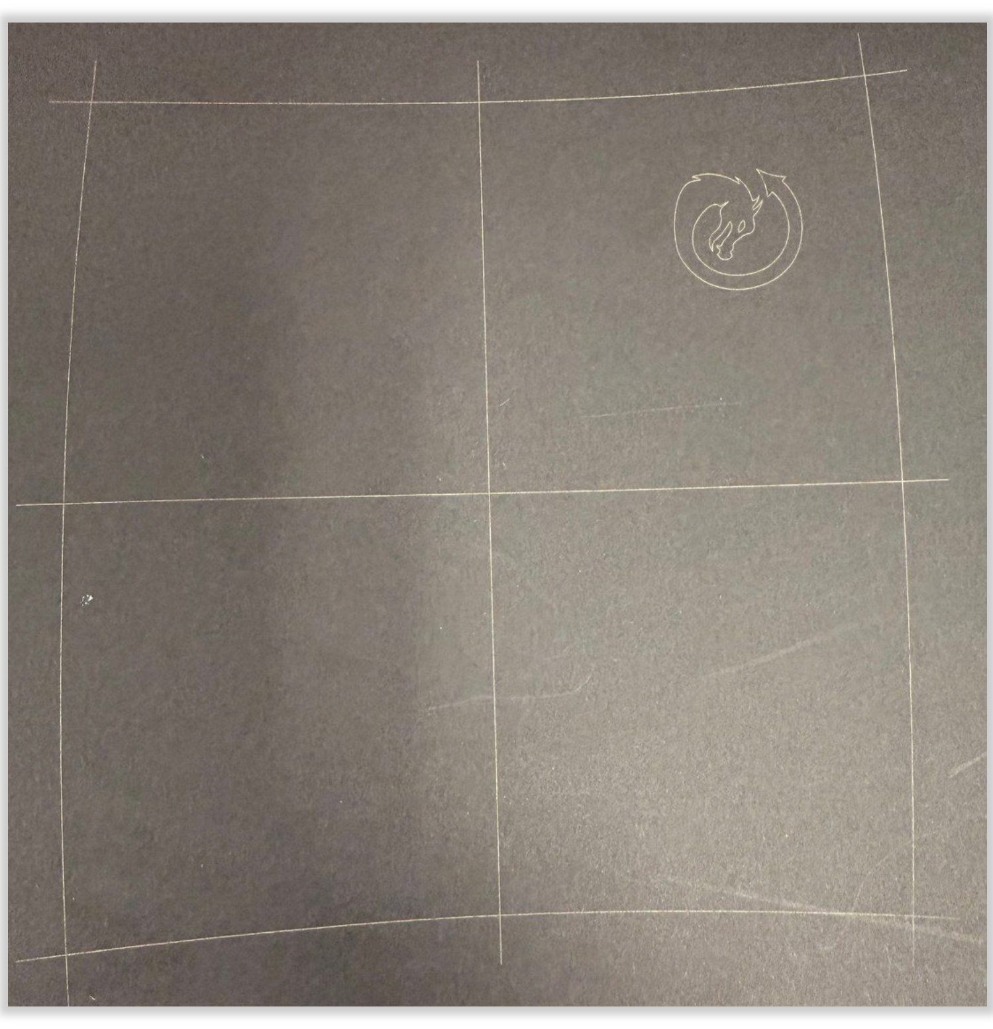
You will need to calibrate the F290 Lens.

To do so, be sure to navigate to the **Laser Tools > Calibrate Galvo Lens > 9 Point Correction (Easy).**

Follow the steps.

⚠ **Important**: You will need dark colored construction paper (we recommend black construction paper) and a caliper.

Note: Lenses can be swapped between the F160 and F290 models without requiring recalibration, as long as the F290 lens is listed under Devices and its calibration has been completed. However, each time a lens is changed, the focal point must be re-established. This involves manually adjusting the set screws until the two red diode dots converge, confirming proper alignment for the installed lens. Refer to Step 3.



This image illustrates the expected reference markings on the black construction paper following execution of the Galvo lens calibration procedure.

Feedback